

Special Notice (SN) DARPA-SN-26-47
Automated Discovery for Design and Control of Turbulent Systems (AutoDIDACTS)
Proposers Day
April 1, 2026

The Defense Advanced Research Projects Agency (DARPA) Defense Sciences Office (DSO) is sponsoring a Proposers Day webcast to provide information to potential proposers on the objectives of an anticipated Broad Agency Announcement (BAA) for the Automated Discovery for Design and Control of Turbulent Systems (AutoDIDACTS) program. The Proposers Day will be held via webcast on April 1, 2026, from 10:00 a.m. to 5:00 p.m. (Eastern Time). Advance registration is required for viewing the webcast. Note that all times listed in this announcement and on the registration website are Eastern Time.

The goals of the AutoDIDACTS Proposers Day are to: (1) introduce the research community (industry, academia, and Government) to the AutoDIDACTS program vision and goals; (2) explain the mechanics of a DARPA program and the milestones of this particular effort; and (3) encourage and promote teaming arrangements among potential organizations that have the relevant expertise, facilities, and capabilities for executing a research and development program responsive to the AutoDIDACTS program goals.

DARPA anticipates releasing the AutoDIDACTS BAA in March 2026. If released, the BAA will be available on <https://sam.gov/> and <http://www.grants.gov/>. Following Proposers Day, DARPA may post the presented materials to the DARPA website.

To maximize the pool of innovative proposal concepts, DARPA strongly encourages participation by non-traditional proposers (including small businesses, academic and research institutions, and first-time government contractors) in events such as this and any subsequent solicitation.

PROGRAM OBJECTIVE AND DESCRIPTION

The AutoDIDACTS program aims to develop and validate new paradigms for rapid design exploration and innovation across regimes of aeronautical design and turbulence control problems of interest to the Department of War (DoW).

Despite the growing sophistication of computational tools for aeronautical design, the rapid exploration of design space is limited by both the accuracy and generalizability of computational models. As has been observed across regimes of aeronautics—from low-Reynolds number design of micro-air vehicles (MAVs), to predictive modeling of transonic and unsteady flight regimes, to multi-objective optimization in supersonic and extreme conditions—a common underlying limitation has been the lack of accurate, predictive, and generalizable macroscale design principles that can guide design exploration and innovation.

Building upon recent successes in the data-informed development of predictive and generalizable turbulence models, the AutoDIDACTS program is focused on the automated data-informed discovery of universal and predictive macroscale principles for design and control of turbulent systems. In essence, the principal objective of this program is to demonstrate the use of precision ground-truth data to go beyond microscopic models of turbulence closures and to arrive directly at macroscale principles that can inform aeronautical design in various regimes of relevance to the DoW. Of particular interest to this program is the development of data-informed

design principles and models that combine high predictive accuracy with generalizability across wide parameter regimes, multi-objective and multi-parameter design optimization, and inverse design paradigms.

This program will require teams to combine expertise in (1) compact and tunable experimental simulators of turbulent flow phenomenology for high-throughput generation of precision ground-truth data; (2) development of data-informed techniques for discovery of macroscale predictive principles and surrogate models; and (3) design and development in the relevant topic areas for demonstrations of end-of-program challenge problems (see below).

Successful teams will validate their approach during Phase I of the program via performer-defined benchmarks and via comparisons of their developed models to independent computational fluid dynamics (CFD) computations and measurements in state-of-the-art wind-tunnels. Teams that succeed in such validation of design principles and predictive surrogate models may progress to specific challenge problems whose metrics will be released in a forthcoming solicitation. Topic areas of these challenge problems may include, but are not limited to, design and demonstration of high-endurance MAVs; demonstration of adaptive control techniques and low-latency fly-by-wire systems for transitional or unsteady conditions; multi-objective optimization for supersonic and extreme conditions; and development of optimal protocols for resilient optical communications, imaging, and time synchronization in turbulent atmospheric conditions.

TEAMING

DARPA highly encourages teaming before proposal submission and will facilitate forming teams with the necessary expertise. Potential proposers may choose to participate in either, none, or both of the following options:

1. Attendee List (limited distribution): Participant contact information (name, organization, email address) will be distributed to all other AutoDIDACTS Proposers Day attendees. The registration website will ask registrants to indicate whether they approve distribution of their contact information.
2. Proposer Profile List (limited distribution): Interested parties will submit a one-page profile consisting of their contact information (name, organization, email, telephone number, mailing address, and, if applicable, organization website), a brief description of their technical competencies, and, if applicable, their desired expertise from other teams/organizations. All profiles must be emailed to AutoDIDACTS@darpa.mil no later than 4:00 p.m. on April 3, 2026. Following the deadline, the consolidated teaming profiles will be sent via email to the proposers who submitted a profile. Profile content, communications, networking, and team formation are the sole responsibility of the proposers. Neither DARPA nor the DoW endorses the information and organizations contained in the consolidated teaming profile document, nor does DARPA or DoW exercise any responsibility for improper dissemination of the teaming profiles.

LIGHTNING TALKS

Attendees may be afforded the opportunity to give a Lightning Talk (a brief, 5-minute oral presentation) during the Proposers Day Webcast outlining their interests and capabilities. The purpose of these presentations is to facilitate teaming discussions among the attendees. Upon registering, attendees may indicate if they would like to give a Lightning Talk. Due to limited availability, DARPA will accept submissions on a first-come, first-served basis and does not

guarantee that an attendee's requests to brief will be fulfilled. Submitted briefing materials are limited to a single, PDF-format slide, which should be appropriate for public release, as they will be shared with the session via webcast. The slide must be submitted to AutoDIDACTS@darpa.mil by 4:00 p.m. on March 27, 2026. DARPA will contact submitters upon receipt of their slide with additional guidance for the webcast.

SIDEBARS

There will be an opportunity to meet individually with the Program Manager, Mukund Vengalattore. Eight-minute, individual, virtual meetings will be held during Proposers Day between **3:00 p.m. to 5:00 p.m.** Please request an individual/team (limited to four (4) people per team with up to two (2) people per institution) session via the registration website. Requests must be received no later than the registration deadline date. Individual meetings will be scheduled on a first-come, first-served, and space-available basis and limited to two (2) people per institution.

From 3:00 p.m. to 4:00 p.m., the DARPA Contracts Management Office will host a separate meeting targeting proposer contracting personnel to provide more details regarding proposal submission requirement, answer any questions pertaining to contracting and pricing, and provide additional information on how to propose to the anticipated solicitation.

REGISTRATION INFORMATION

The AutoDIDACTS Proposers Day will be webcast on April 1, 2026, from 10:00 a.m. to 5:00 p.m. There is no registration fee for the Proposers Day webcast.

Registration opens: As of publication of this announcement.

Registration website: <https://events.sa-meetings.com/AutoDIDACTSProposersDay>

Registration closes: March 18, 2026, at 4:00 p.m. or when capacity is reached, whichever comes first.

Advance registration via the above website is required in order to receive access to the webcast and is mandatory for every individual intending to view the webcast alone or as part of a group.

Registrants in excess of the maximum capacity limitations (350 individuals) may be added to a waitlist. Individuals who are unable to register because the deadline has passed may request to be added to the waitlist. If slots become available due to cancellations, the slots may be filled on a first come, first served basis from the waitlist.

DARPA encourages participants to register a contracting representative to attend the Proposers Day and participate in the separate meeting with the DARPA Contracts Management Office from 3:00 p.m. to 4:00 p.m.

ELIGIBILITY

DARPA hosts Proposers Days to: (1) promote teaming arrangements between researchers; (2) provide potential proposers with information on whether and how they might respond to the government's research and development solicitations; and (3) increase efficiency in proposal preparation and evaluation. Therefore, Proposers Days are only open to registered potential proposers, i.e., the events are closed to the general public and media. For this particular program, Proposers Day registration is open to potential proposers who are U.S. citizens, U.S. permanent residents, and foreign nationals.

All registrants who are not U.S. citizens must complete and submit either a DARPA Form 60 (U.S. Permanent Resident and Foreign National Visit Request – e.g., industry or academia) or an Official Visit Request (foreign government personnel, only) through their country’s embassy based in Washington, DC, no later than 4:00 p.m. on March 18, 2026. Form 60 submission instructions are provided on the registration website and in the registration confirmation email. Registrants who have submitted a Form 60 that was approved within the last six months do not need to re-submit. The Form 60 will remain valid for six months. Contact your Embassy staff for assistance in submitting the Official Visit Request.

Q & A

Administrative, technical, and contractual questions about the AutoDIDACTS should be emailed to AutoDIDACTS@darpa.mil. All questions must be in English and must include the name and email address of a point of contact. Following Proposers Day, DARPA may post an FAQ list on the DARPA website.

CONTACT INFORMATION

Please refer to the AutoDIDACTS Proposers Day in all correspondence.

- Technical Point of Contact: Dr. Mukund Vengalattore, Program Manager, AutoDIDACTS@darpa.mil
- Registration Website: <https://events.sa-meetings.com/AutoDIDACTSProposersDay>
- DARPA Opportunities Website: <https://www.darpa.mil/research#research-opportunities>

DISCLAIMERS AND IMPORTANT NOTES

Viewing the AutoDIDACTS Proposers Day webcast is voluntary and is not required to propose to the anticipated AutoDIDACTS BAA. Interested parties to this notice are cautioned that nothing herein obligates the Government to issue a solicitation. The Proposers Day does not constitute a formal solicitation for proposals or abstracts. This announcement is issued solely for information and program planning purposes and is not a Request for Information (RFI). Since this is not an RFI, the Government will not accept submissions against this notice, with the exception of the one-page profile referenced in the Teaming section above and briefing materials associated with Lightning Talks. No classified, International Traffic in Arms Regulations (ITAR) controlled, or proprietary information shall be presented by the Government at Proposers Day. Including proprietary information during sidebars or Lightning Talks is at the presenter’s discretion; however, any submitted Lightning Talk presentation slides should be appropriate for public release. DARPA will not provide reimbursement for costs incurred to participate in this Proposers Day. Recording Proposers Day webcast is not permitted.

For those new to DARPA or national security, DARPA makes available a free, comprehensive resource via DARPAConnect on how to do business with the agency. In addition to DARPA 101 materials, relevant preparatory modules include “Making the Most of a Proposers Day” and “Understanding DARPA Broad Agency Announcements.” Registration and access are free at www.darpaconnect.us.